



Optimizing Dye Removal from Water by Coupling Mg–Al Layered Double Hydroxide Adsorption with UV/H₂O₂ Oxidation

R. Devesa-Rey · P. Hidalgo · E. Arce[✉] ·
L. Gonzalez-Gil

Received: 17 April 2026 / Accepted: 23 August 2026
© The Author(s) 2026

Abstract Textile dye removal remains challenging because conventional wastewater treatment plants are not specifically designed to eliminate persistent colorants. In this study, calcined Mg–Al layered double hydroxide (LDH) adsorption, UV irradiation, and H₂O₂ oxidation were evaluated individually and in combination for the removal of two model textile dyes with different ionic characteristics: methyl orange (MO, anionic) and crystal violet (CV, cationic). The coupled treatment was optimized using a Box–Behnken experimental design and a hierarchical reduced polynomial model including selected mixed third-order terms. Among the individual treatments, LDH adsorption showed the highest removal

efficiency. For MO, 99% removal was achieved within 20 min using 2.5 mg mL⁻¹ LDH. In contrast, CV adsorption was slower and required 12 h and 5 mg mL⁻¹ LDH to reach 99% removal. UV irradiation and H₂O₂ oxidation applied separately showed limited effectiveness, with removal efficiencies below 30% after 1 h for both dyes. The hierarchical reduced polynomial model obtained for MO and CV showed coefficients of determination of 0.9630 and 0.9987, respectively, and were used to identify the optimal treatment conditions. For MO, removal above 99% was achieved using 3.2 mg mL⁻¹ LDH and 2 h of UV irradiation without H₂O₂. A comparable removal efficiency could also be achieved using 4.2 mg mL⁻¹ LDH alone, confirming that adsorption was the dominant removal mechanism for this anionic dye. In contrast, near-complete CV removal required the combined application of 8.7 mg mL⁻¹ LDH, 32 min of UV irradiation, and 5.7% H₂O₂ stock solution. A generalized condition consisting of 4.5 mg mL⁻¹ LDH, 1 h of UV irradiation, and 3.5% H₂O₂ stock solution achieved removal efficiencies above 98% for both dyes. The LDH also maintained MO removal efficiencies above 90% over ten consecutive reuse cycles and achieved 94% removal after thermal regeneration. Overall, these findings highlight the potential of the LDH/UV/H₂O₂ system as a versatile treatment strategy for dye-contaminated water, with its performance strongly influenced by the ionic characteristics of the target dye.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11270-026-09897-0>.

R. Devesa-Rey (✉) · L. Gonzalez-Gil
Defense University Centre at the Spanish Naval Academy,
Group of Thermal and Environmental Engineering
(InTeam), Plaza de España, 36920 Marín, Pontevedra,
Spain
e-mail: rosa.devesa.rey@ud.uvigo.es

P. Hidalgo
Spanish Army, Ministry of Defense, 28046 Madrid, Spain

E. Arce (✉)
Department of Industrial Engineering, Polytechnic
School of Engineering of Ferrol, University of A Coruña,
15403 Ferrol, A Coruña, Spain
e-mail: elena.arce@udc.es

Keywords Advanced oxidation process · Dyes · Hydrotalcite · Box-Behnken design · Textile wastewater

1 Introduction

The textile industry follows a linear economic model, resulting in high costs and the consumption of economic and natural resources. It is considered the second most polluting industry worldwide, after the petroleum industry, accounting for 20% of global wastewater generation and 8% of carbon dioxide emissions into the atmosphere (Urbina-Suarez et al., 2024). Environmental concerns related to the textile industry are mainly associated with wastewater discharge and pollution, as well as with textile waste recycling. There is worldwide concern about the need to implement sustainable practices in order to reduce the impact of textile waste.

The textile industry is known for its considerable water use and the subsequent generation of wastewater (Bora et al., 2026). One possible reason is the high level of clothing consumption associated with the paradigm known as "fast fashion" (Klaka & Schlittmeier, 2026) which involves the production of a large volume of low-quality clothing and the extensive use of dyes, resulting in increased use of toxic substances over shorter timeframes (Pinto & Peleg Mizrahi, 2025). This evolution has also led to the development of new dyes for synthetic fibers such as polyester, polyamide, nylon, and elastane (Stanton et al., 2023). In parallel, the use of chemical reagents such as acids, sulfates, and phenols for processing synthetic fibers has also increased (Rajabinejad et al., 2019). The main challenge faced by the textile industry today lies in reducing the waste generated and the chemicals used, which pose significant threats to ecosystems. In particular, textile dyes may act as skin allergens, endocrine disruptors, or respiratory allergens, and some may exhibit carcinogenic, mutagenic and reproductive properties (Bour et al., 2023).

This study arises from the crucial need to explore novel methods for enhancing the management of wastewater generated by the textile industry, specifically focusing on the removal of dyes discharged into the environment. Conventional methods are currently expensive and insufficiently efficient (Senthil Rathi & Senthil Kumar, 2022), since these compounds exhibit

remarkable persistence (Teow et al., 2023). Consequently, there is a pressing demand for technologies that facilitate their removal, enabling the reuse of substantial volumes of water consumed during industrial processes. Current research is focused on developing new treatment systems that are easier to apply and exhibit superior contaminant removal performance, particularly for dyes present in water. Dye-contaminated water can be treated through adsorption, photolysis, advanced oxidation, photocatalysis, and combinations of these processes.

Among adsorption materials, layered double hydroxides (LDHs) are a class of layered materials whose structure consists of cationic layers intercalated with exchangeable anions and hydrated species. A typical LDH consists of divalent and trivalent metal cations, mostly belonging to the third and fourth periods of the periodic table (such as Mg^{2+} , Al^{3+} , Zn^{2+} , Fe^{3+}). The chemical formula of an LDH is expressed as $[M_{1-x}^{2+}M_x^{3+}(OH)_2]^{x+}(A^{n-})_{x/n} \cdot nH_2O$, where M^{2+} and M^{3+} are cations located in the hydroxide layers and A^{n-} is the interlayer anion (Gupta et al., 2024). LDHs have attracted considerable attention in wastewater treatment due to their high surface area, variable composition, and versatile surface chemistry, which allow them to effectively remove a wide range of contaminants. In particular, LDHs can exchange anions in their interlayer spaces, making them effective for removing anions such as phosphates (Hassanzadeh et al., 2024), nitrates (Díaz et al., 2024), and sulfates (Masoud et al., 2024) from wastewater. The benefits of employing LDHs in wastewater treatment include their relatively low-cost synthesis and their ability to create specific environments for retaining a wide range of contaminants. Although their use has typically focused on the adsorption of anionic contaminants (Devesa-Rey et al., 2021), their versatile surface chemistry has also been employed to remove cationic pollutants, such as lithium ions, heavy metals and organic compounds (Li et al., 2025; Su et al., 2024; Qi et al., 2024), among others. Among these contaminants, dyes have received particular attention, although the reported removal efficiencies vary considerably depending on the dye, LDH composition, and wastewater matrix. Reported efficiencies include 85–91% for methylene blue depending on the wastewater matrix (Munonde et al., 2025), up to 100% for crystal violet (CV) and 79% for methyl orange (MO) using CuFe-LDH membranes (Zubair et al., 2024),

and 99.49% for methyl orange using polyethylene glycol-modified LDH, although reuse was limited to three cycles (Kameliya et al., 2024).

Recent studies have demonstrated that the functionalization of adsorbent materials can enhance their affinity, selectivity, and recovery capacity for different aqueous pollutants, including dyes and metal ions (Kubra et al., 2021; Hasan et al., 2023). In particular, polymeric and ligand-functionalized composites have achieved high removal efficiencies through the incorporation of specific binding sites, although their preparation may involve multiple synthesis steps and additional chemical reagents (Awual et al., 2023; Salman et al., 2023). Chitosan-based nanocomposites have also shown promising performance for dye removal, confirming the potential of surface modification and adsorption-process optimization for treating dye-contaminated water (Waliullah et al., 2023). Accordingly, the interest of calcined Mg–Al LDH lies not in assuming universal superiority over previously reported materials, but in evaluating its performance as a relatively simple adsorbent.

In addition to LDH-based adsorption, UV/H₂O₂ oxidation, LDH-assisted advanced oxidation, and coupled adsorption–photocatalytic processes have been investigated for dye treatment, because dye properties, including their ionic nature, can strongly influence their response to different treatment processes. Representative studies are compared in Table 1, considering the initial dye concentration, material and oxidant dosages, treatment time, removal efficiency, evidence of mineralization or transformation, adsorbent reusability, and validation in real wastewater. Although near-complete dye removal has been reported using several of these approaches, direct comparisons remain difficult because treatment performance depends strongly on the dye and its initial concentration, material and oxidant dosages, irradiation conditions, analytical endpoint, regeneration procedure, and water matrix. Moreover, systematic studies comparing LDH adsorption, UV irradiation, and H₂O₂ treatment individually and in combination under the same experimental conditions remain limited, particularly for dyes with different ionic characteristics.

To the best of our knowledge, no previous study has systematically compared and optimized calcined Mg–Al LDH adsorption, UV irradiation, and H₂O₂ treatment, both individually and in combination,

within the same experimental framework for methyl orange and crystal violet. Accordingly, the specific objectives of this study were to: (i) systematically compare the individual and combined performance of LDH adsorption, UV irradiation, and H₂O₂ treatment; (ii) evaluate two model dyes with contrasting ionic and structural characteristics to determine how the most suitable treatment route depends on the properties of the dye; (iii) apply a Box–Behnken experimental design to identify both maximum-removal conditions and alternative operating combinations within the investigated experimental domain; and (iv) assess the reusability of the Mg–Al LDH for methyl orange adsorption over consecutive cycles.

2 Materials and Methods

2.1 Dye Selection and Measurement

Two synthetic dyes were studied, methyl orange (MO) and crystal violet (CV), both purchased from Sigma Aldrich (ACS-grade reagents). These dyes, with reported textile applications, are frequently used as model anionic azo and cationic contaminants, respectively, are persistent in the environment, and have documented harmful effects on living organisms and human health (Kishor et al., 2021; Mani & Bhargava, 2016). To determine their maximum absorption wavelengths, spectral scans from 200 to 950 nm were performed, and the maximum absorption peaks were identified at 465 nm for methyl orange and 590 nm for crystal violet. Their main chemical characteristics are summarized in Table 2. The concentrations used in all the treatments were 20 mg·L⁻¹ for methyl orange and 10 mg·L⁻¹ for crystal violet, which are within the typical range reported for dyes in textile wastewater (Behera et al., 2021). The concentrations of both dyes were determined by analyzing aliquots collected during the experiments using a UV–VIS spectrophotometer (Lambda XLS, Perkin Elmer).

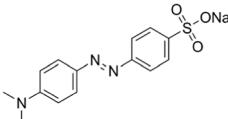
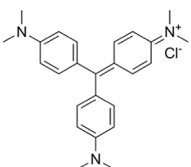
The decrease in absorbance was interpreted as evidence of decolorization and apparent dye removal from the aqueous phase. Under adsorption conditions, this response is consistent with the transfer of the intact dye to the Mg–Al LDH, whereas under UV/H₂O₂ treatment it may also reflect chemical transformation of the chromophoric structure. However, single-wavelength absorbance measurements do not

Table 1 Comparison of representative LDH adsorption, LDH-assisted advanced oxidation, UV/H₂O₂ and coupled adsorption–photocatalysis processes for dye removal

Study	Treatment and adsorbent/catalyst	Dye and initial concentration	Adsorbent/catalyst and oxidant dosage	Reaction time	Maximum dye removal	TOC/COD removal or other degradation evidence	Reusability
(Kameliya et al., 2024)	Adsorption using PEG-400-modified Mg–Al LDH	Methyl orange; 100 mg L ⁻¹	0.5 g L ⁻¹ LDH	20 min	99.49%	NR	Three cycles; 88.23% MO recovery after the third cycle
(Haji et al., 2011)	UV/H ₂ O ₂ advanced oxidation	Methyl orange; 0.078 mM (approximately 25.5 mg L ⁻¹)	0.1 mL H ₂ O ₂ solution added to 20 mL dye solution	3 min	Complete decolorization	NR	Not applicable
(Munonde et al., 2025)	Adsorption followed by visible-light photocatalysis using NiFe-LDH supported on tyre-ash-derived activated carbon	Methylene blue; 30 mg L ⁻¹ ; real textile wastewater: 2.33–6.17 mg L ⁻¹	30 mg catalyst in 100 mL (0.30 g L ⁻¹)	30 min adsorption followed by 30 min visible-light irradiation	Model solution: 94.2%; textile wastewater: 85–87%; river water: 90–92%	TOC/COD: NR; transformation products analyzed by UPLC–MS; radical-scavenging tests conducted	Five cycles; removal efficiency decreased by 20.8%
(Zubair, et al., 2024)	PVDF-supported CuFe-LDH membrane combined with UV-LED/persulfate oxidation	Methylene blue; 29.45–101.93 mg L ⁻¹ in optimization experiments; CV, MO, and EBT also evaluated	Membrane catalyst: 1.84–1.95 g L ⁻¹ ; persulfate: 0.50–2.41 g L ⁻¹	45.22–178.5 min in optimization experiments; 45 min in comparative dye tests	MB: 100%; CV: 100%; MO: 79%; EBT: 60%	TOC/COD: NR; reactive-species contribution investigated	Five consecutive cycles
(Yadav & Dasgupta, 2022)	Adsorption using nitrate-intercalated Mg–Al LDH	Methyl orange; 30 mg L ⁻¹	10 mg in 50 mL, equivalent to 0.20 g L ⁻¹	60 min	98.22% at 25 °C; maximum adsorption capacity of 583 mg g ⁻¹	NR	Three cycles without a significant decrease in adsorption capacity
(Hamzah et al., 2025)	Adsorption using a palm-oil-fuel-ash/Mg–Fe LDH composite	Reactive Orange 16 and crystal violet; 100 mg L ⁻¹ in kinetic experiments; real wastewater dye concentration NR	1 g L ⁻¹ in model-solution studies; effective dosage of 5 g L ⁻¹ for real textile wastewater	Equilibrium after 90 min for RO16 and 120 min for CV; real-wastewater treatment time NR	Maximum capacities: 444.07 mg g ⁻¹ for RO16 and 1048.69 mg g ⁻¹ for CV; > 90% color removal in real textile wastewater	63% COD and approximately 60% TOC removal from real textile wastewater	Five adsorption–regeneration cycles
(Shu, 2006)	UV/H ₂ O ₂ oxidation and sequential ozonation–UV/H ₂ O ₂ treatment	C.I. Direct Blue 199 and real cotton dye-bath effluent; initial concentrations NR	H ₂ O ₂ dosage 116.32 Mm UV power: 560 W	30 min for DB199; 120 min for real dye-bath effluent	UV/H ₂ O ₂ alone: 80% color removal for DB199 and 95% for the real effluent	Approximately 75% TOC removal for DB199 and 80% for the real dye-bath effluent	Not applicable

CV crystal violet, EBT Eriochrome Black T, H₂O₂ hydrogen peroxide, LDH layered double hydroxide, MB methylene blue, MO methyl orange, NR not reported, PEG polyethylene glycol, PVDF polyvinylidene fluoride

Table 2 Chemical characteristics of the two selected dyes. Data found in PubChem database

Name	Chemical structure	Molecular formula	Chemical nature	CAS number	MW (g/mol)	$\lambda_{\max}$ (nm)	PKa (298 K)
Methyl Orange		$C_{14}H_{14}N_3NaO_3S$	Anionic	547–58-0	327.33	465	3.4
Crystal Violet		$C_{25}H_{30}ClN_3$	Cationic	548–62-9	407.98	590	8.6

unequivocally confirm either the disappearance of the parent dye molecule or its mineralization.

2.2 Synthesis and Characterization of LDH

The LDH was synthesized by co-precipitation. $Mg(NO_3)_2$ and $Al(NO_3)_3$ were used at a 3:1 molar ratio and added to a NaOH solution at a constant rate, under continuous stirring, while the pH was maintained at 10 using a pH meter (HANNA, Edge pH). The resulting suspension was continuously stirred for 4 h. It was then aged for 18 h, filtered, and oven-dried at 100 °C for 24 h. Next, the synthesized LDH was ground to obtain a particle size of 0.5–1.0 mm. Finally, the material was calcined for 2 h at 600 °C.

The LDH was characterized using Brunauer-Emmett-Teller (BET) analysis to determine the surface area, pore volume, and pore size of the adsorbent. The operation conditions were as follows: sample mass: 0.2959 g; cold free space: 47.2417 cm^3 ; analysis adsorptive: N_2 ; analysis bath temperature: 77.138 K; equilibration interval: 10 s; automatic degasification: yes. X-ray diffraction was used to confirm the crystalline structure of the LDH using a D8 Advance X-Ray diffractometer, with a wavelength of 0.1542 nm and 2θ range from 2° to 65°.

2.3 Characteristics of the H_2O_2 Reagent and the UV Reactor

Chemical oxidation was performed using hydrogen peroxide (H_2O_2), purchased from Foret at a

concentration of 30% (v/v). Further details of the composition of this reagent are provided in Table S1. The UV treatment was performed in a stainless-steel reactor, equipped with a low-pressure mercury UV-C lamp (Rotek, Taiwan). The lamp emitted radiation with a peak wavelength of 254 nm, and the UV dose was 30 $mJ\cdot cm^{-2}$. The reactor was operated in batch mode. Additional characteristics of the UV reactor are provided in Table S1.

2.4 Experimental Setup

2.4.1 Preliminary Experiments with Individual Treatments

Preliminary experiments were conducted with two objectives: (i) to evaluate the capacity of LDH, UV irradiation, and H_2O_2 treatment to achieve the apparent removal of methyl orange and crystal violet from the liquid phase when applied individually; and (ii) to define the ranges of the treatment parameters to be used as variables in the Box-Behnken design.

The first step consisted of evaluating the adsorption of methyl orange and crystal violet onto the LDH. For methyl orange, an initial experiment was carried out by mixing 50 mL of dye solution (20 $mg\ L^{-1}$) with 0.250 g of LDH (5 $mg\ mL^{-1}$). The mixture was agitated in an orbital shaker at 200 rpm, and aliquots were taken after 30, 60 min and 12 h to determine color removal. Before the samples were transferred to the spectrophotometer cuvettes, they were filtered through 0.45- μm membrane filters to remove the suspended LDH, thereby ensuring

more stable absorbance values. Subsequently, two additional experiments using 50 mL of methyl orange solution were carried out. One experiment contained 0.125 g of calcined LDH (2.5 mg mL^{-1}), while the other one contained 0.0625 g (1.25 mg mL^{-1}). The same procedure was followed, but samples were taken during the first 20 min at 5-min intervals. Similarly, an initial experiment was carried out with 50 mL of crystal violet solution (10 mg L^{-1}) and 0.250 g of LDH (5 mg mL^{-1}). Then, two additional 50-mL experiments were prepared. One contained 0.125 g of calcined LDH (2.5 mg mL^{-1}), while the other contained 0.0625 g (1.25 mg mL^{-1}). The same procedure used for methyl orange was followed, but aliquots were taken after 20, 40, and 60 min, and a final measurement was performed after 12 h.

Next, the influence of UV irradiation was studied by placing 200 mL of the corresponding dye solution in a batch reactor equipped with a UV lamp and irradiating it for 12 h. Aliquots were taken after 30 and 60 min and 12 h to determine the apparent dye concentration. Finally, the influence of H_2O_2 treatment was studied in 25-mL experiments containing each dye solution and 5% (v/v) of the commercial 30% H_2O_2 solution. The experiments were agitated at 200 rpm, and samples were taken after 30 and 60 min. All preliminary experiments were carried out in triplicate at room temperature.

2.4.2 Combined Treatment for Color Removal:

Box-Behnken Design

The combined effects of LDH dosage, UV irradiation time, and H_2O_2 dosage were evaluated using a three-factor, three-level Box–Behnken design comprising 15 experimental runs, including three replicates at the center point (Box & Behnken, 1960). The response variable was the dye-removal efficiency expressed as a percentage. The independent variables were expressed in coded form according to:

$$x_i = \frac{X_i - X_{0,i}}{\Delta X_i} \quad (1)$$

where x_i is the coded value of factor i , X_i is its actual value, $X_{0,i}$ is the value at the center point, and ΔX_i is the step change between coded levels. For methyl orange, the coded variables were calculated as follows:

$$x_1 = \frac{C_{\text{LDH}} - 2.5}{2.5}, x_2 = \frac{t_{\text{UV}} - 1}{1}, x_3 = \frac{C_{\text{H}_2\text{O}_2} - 5}{5} \quad (2)$$

where C_{LDH} is expressed in mg mL^{-1} , t_{UV} in h, and $C_{\text{H}_2\text{O}_2}$ as the volumetric percentage of the H_2O_2 stock solution.

For crystal violet, the coded variables were:

$$x_1 = \frac{C_{\text{LDH}} - 5}{5}, x_2 = \frac{t_{\text{UV}} - 30}{30}, x_3 = \frac{C_{\text{H}_2\text{O}_2} - 5}{5} \quad (3)$$

where t_{UV} is expressed in min.

The initial regression analysis considered linear and quadratic terms, two-factor linear interactions, and selected estimable mixed third-order terms. The augmented empirical model was expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{33} x_3^2 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{122} x_1 x_2^2 + \beta_{112} x_1^2 x_2 + \beta_{113} x_1^2 x_3 + \varepsilon \quad (4)$$

where y is the predicted dye-removal efficiency, β_0 is the intercept, β_i are the linear coefficients, β_{ii} are the quadratic coefficients, β_{ij} are the two-factor interaction coefficients, β_{ijk} are the coefficients of the selected mixed third-order terms, and ε is the residual error.

Reduced empirical models were obtained by removing non-significant terms, except for lower-order terms retained to preserve model hierarchy. The resulting models were subsequently refitted using least-squares regression. Model adequacy was evaluated by analysis of standard error, t and p values, R^2 and adjusted R^2 and 95% confidence intervals. Agreement between the predicted and observed responses was also assessed using predicted-versus-observed plots.

The experimental data were analysed using response-surface methodology with Statistica 7.0 software. Based on the preliminary results, a three-factor, three-level incomplete factorial design comprising 15 experiments was established, in which the LDH and hydrogen peroxide concentrations and the UV irradiation time were varied. The experimental design included three replicates at the center point (experiments 13–15), which were performed at the central level of each independent variable to assess experimental variability.

The experimental conditions used for the independent variables, expressed as coded levels, are shown in Table 3. The standardized, dimensionless

Table 3 Independent and dependent variables analyzed in the methyl orange (MO) and crystal violet (CV) color reduction

INDEPENDENT VARIABLES			
Variable	Nomenclature	Units	Range of variation
LDH concentration	[LDH]	mg mL ⁻¹	[0, 5] (MO) [0, 10] (CV)
UV exposure time	[UV]	hours	[0, 2] (MO) [0, 1] (CV)
H ₂ O ₂ dose	[H ₂ O ₂]	%	[0, 10] (MO, CV)
ADIMENSIONAL CODED VARIABLES			
Variable	Nomenclature	Units	Range of variation
Dimensionless LDH concentration	x_1	([LDH]–2.5)/2.5 (MO) ([LDH]–5)/5 (CV)	[–1, 1]
Dimensionless UV exposure time	x_2	([UV]–1)/1 (MO) ([UV]–0.5)/0.5 (CV)	[–1, 1]
Dimensionless H ₂ O ₂ dose	x_3	([H ₂ O ₂]–5)/5 (MO, CV)	[–1, 1]
DEPENDENT VARIABLES			
Variable	Nomenclature	Units	
Color removal (methyl orange, 465 nm)	y_1	%	
Color removal (crystal violet, 590 nm)	y_2	%	

coded variables, ranging from –1 to +1, were defined as x_1 (LDH dosage), x_2 (UV irradiation time) and x_3 (H₂O₂ concentration). The dependent variables were the color- decolorization efficiencies measured at 465 nm for methyl orange and 590 nm for crystal violet, corresponding to variables y_1 and y_2 , respectively.

The ranges selected for the independent variables were based on the results obtained in the preliminary experiments. The selected ranges were intended to allow potential improvements in decolorization efficiency resulting from the combined treatments to be detected. Experiments performed in the absence of UV irradiation were conducted for 10 min under stirring at 200.

2.5 Adsorption Kinetic Analysis

The time-dependent adsorption data obtained during the preliminary LDH experiments were additionally analyzed using pseudo-first-order (PFO), pseudo-second-order (PSO), and Weber–Morris intraparticle-diffusion (IPD) models. The amount of dye adsorbed at a given time, q_t (mg g⁻¹), was calculated as:

$$q_t = \frac{(C_0 - C_t)V}{m} \quad (5)$$

where C_0 and C_t are the initial and time-dependent dye concentrations (mg L⁻¹), respectively, V is the solution volume (L), and m is the mass of LDH (g).

The PFO and PSO models were fitted by nonlinear regression according to:

$$q_t = q_e(1 - e^{-k_1 t}) \quad (6)$$

and

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t} \quad (7)$$

respectively, where q_e is the fitted adsorption capacity at equilibrium, k_1 is the PFO rate constant (min⁻¹), and k_2 is the PSO rate constant (g mg⁻¹ min⁻¹). The possible contribution of intraparticle diffusion was evaluated using the Weber–Morris relationship:

$$q_t = k_{id} t^{1/2} + C \quad (8)$$

where k_{id} is the intraparticle-diffusion constant and C is the intercept. Model performance was compared using the coefficient of determination (R^2) and root-mean-square error (RMSE). For crystal violet, kinetic fitting was restricted to the measurements obtained between 0 and 60 min. The measurement obtained

after 12 h was excluded from kinetic fitting because the sample was maintained under sedimentation during this period and therefore did not correspond to the same hydrodynamic conditions.

2.6 Reuse and Recovery of LDH

The reusability of the LDH was evaluated by conducting ten consecutive experiments using an LDH dosage of 2.5 mg mL^{-1} to treat 20 mg L^{-1} of methyl orange solution for 20 min. Reuse experiments were conducted with methyl orange because preliminary adsorption tests showed substantially faster and stronger uptake by the Mg–Al LDH than for crystal violet, making methyl orange suitable for evaluating repeated adsorption cycles. After each experiment, the treated solution was separated from the LDH, and the adsorbent was directly reused in the next experiment. After ten consecutive adsorption cycles, the recovered LDH was thermally regenerated once at 600°C for 30 min and subsequently tested in an additional methyl orange adsorption experiment.

3 Results and Discussion

3.1 Characterization of LDH

LDHs are anionic clays characterized by positively charged metal hydroxide layers separated by inter-layer anions and water molecules. BET analysis is used to evaluate the physical adsorption of gas molecules onto the solid surface of the LDH and to determine its specific surface area. Adsorption isotherms are obtained by plotting the amount of gas adsorbed against the relative pressure and assuming a multi-layer adsorption process to calculate the surface area (Fripiat et al., 1986). In contrast, the determination of the Langmuir surface area assumes monolayer adsorption on a surface with uniform adsorption energies. Therefore, BET measurements are generally more applicable to porous materials such as LDHs.

In this study, the BET and Langmuir surface areas were $59.87 \text{ m}^2 \text{ g}^{-1}$ and $82.70 \text{ m}^2 \text{ g}^{-1}$, respectively. The LDH synthesized in this study by the coprecipitation method had a surface area similar to values reported in other studies (Zhou et al., 2021) and higher than those reported for Mg–Al LDH synthesized using other methodologies, such as hydrothermal synthesis,

for which values ranging from 19.21 (Kameliya et al., 2024) to $51.91 \text{ m}^2 \text{ g}^{-1}$ (Kishor et al., 2021) have been reported. However, other authors have found considerable variability in the surface areas of LDHs synthesized under similar conditions, with values ranging from 47.9 to $106 \text{ m}^2 \text{ g}^{-1}$ (Wijaya et al., 2023). This variability highlights the importance of thoroughly characterizing the surface area because of its influence on the adsorption capacity of LDHs.

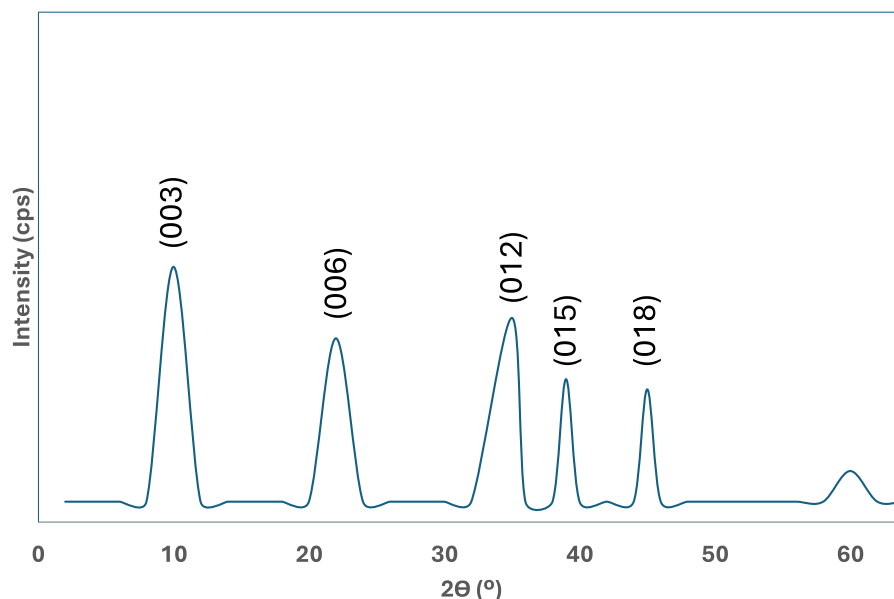
The N_2 adsorption-desorption isotherm of the LDH was compared with the isotherms defined by the International Union of Pure and Applied Chemistry (IUPAC) to identify the predominant pore structure of the material (Fig. S1). The results indicate a type V isotherm, characteristic of mesoporous materials. The hysteresis loop was classified as type H3, which is typical of aggregates of plate-like particles, such as those found in LDH.

The XRD pattern is shown in Fig. 1. The Mg–Al LDH exhibited characteristic peaks corresponding to the (003), (006), (012), (015) and (018) planes, confirming the LDH structure according to values reported in the literature (Wang et al., 2017). Despite being heated to 600°C , the Mg–Al LDH sample retained some crystalline features, with clear diffraction lines indicating partial structural stability at high temperatures.

The BET and XRD analyses provide information on the textural properties and crystalline structure of the synthesized Mg–Al LDH. The measured specific surface area ($59.87 \text{ m}^2 \text{ g}^{-1}$) and pore volume ($0.448 \text{ cm}^3 \text{ g}^{-1}$) indicate the presence of an accessible porous structure that can facilitate contact between the adsorbent and dissolved dye molecules. However, these analyses do not provide direct information on the surface charge, specific functional groups involved in adsorption, or changes in the LDH surface after dye uptake. Therefore, the present characterization supports the physical suitability of the material for adsorption but does not allow the molecular adsorption mechanism to be directly established.

The pore-size distribution and cumulative pore surface area were estimated using the Barrett-Joyner-Halenda (BJH) method, which analyzes pore sizes ranging from mesopores, with diameters between $17.0 \text{ \AA}$ and $500 \text{ \AA}$, to macropores, with diameters greater than $500 \text{ \AA}$ and up to $3000 \text{ \AA}$. Mesopores are intermediate-sized pores that facilitate adsorption

Fig. 1 X-ray diffraction (XRD) pattern of the synthesized Mg–Al LDH



because their dimensions are suitable for molecular diffusion and interaction. Macropores, in contrast, facilitate the transport of larger molecules and are typically found in materials requiring high permeability and low resistance to fluid flow. In this study, the BJH cumulative adsorption and desorption surface areas were 70.38 and 130.12 m² g^{−1}, respectively, suggesting that a substantial proportion of the surface area was associated with mesopores. The difference between the adsorption and desorption surface areas indicates a hysteresis effect, which is common in mesoporous materials and suggests the presence of a complex pore structure with a high internal surface area. A high mesopore surface may favor the adsorption of small molecules, while the presence of macropores may facilitate the transport of larger molecules, improving the performance of the LDH in adsorption processes.

Finally, single-point adsorption-desorption analysis was used to determine the total pore volume close to saturation ($P/P_0 \approx 1$). The amount of nitrogen adsorbed at this point was converted into pore volume. The LDH exhibited a pore volume of 0.448 cm³ g^{−1}. Pore volume can vary widely depending on the material. Porous materials such as activated carbons, zeolites, and mesoporous silica typically exhibit pore volumes ranging from 0.1 to 2.0 cm³ g^{−1}. Reported ranges include 0.2 to 1.2 cm³ g^{−1} for activated carbon (Asamoah et al., 2025; Khuong

et al., 2022), 0.2 to 1.5 cm³ g^{−1} for mesoporous silica (Khushnood et al., 2024; Usgodaarachchi, 2021), and 0.1 to 0.8 cm³ g^{−1} for zeolites (Meiramkulova et al., 2024).

With respect to LDHs, previous studies have reported pore volumes ranging from 0.2 to 0.9 cm³ g^{−1}, depending on their layered structure, synthesis method (Bai et al., 2024), and the presence of exchangeable ions (Zhang et al., 2024). The pore volume of 0.448 cm³ g^{−1} obtained for the LDH synthesized in this study falls within this expected range, indicating a moderate pore volume suitable for adsorption applications. This value supports the potential use of the synthesized LDH in adsorption processes and is comparable to values reported for other well-known porous materials.

The measured surface area, pore volume, and mesoporous structure provide accessible adsorption sites and may facilitate the diffusion of dye molecules through the LDH particles. However, these textural properties alone cannot explain the markedly different behavior of methyl orange and crystal violet. Since both dyes were treated with the same material, their contrasting removal rates indicate that adsorption was also strongly influenced by the chemical characteristics and ionic charge of the dye molecules. Therefore, the adsorption performance of the synthesized LDH should be interpreted as the combined result of its porous structure, surface

chemistry, and potential electrostatic interactions with the target dye.

3.2 Color Removal in Individual Experiments by Means of LDH, UV Light and H₂O₂

3.2.1 Dyes Adsorption onto LDH

The results obtained in the preliminary experiments conducted to evaluate the adsorption capacity of the synthesized LDH are shown in Fig. 2. For methyl orange, the first experiment was conducted using an LDH dosage of 5 mg mL⁻¹. The first measurement was taken after 30 min, when almost complete removal (99%) was achieved, with an adsorption rate of 0.132 mg dye g⁻¹ LDH min⁻¹. Subsequent experiments were therefore conducted at lower LDH dosages of 2.5 and 1.25 mg mL⁻¹, and aliquots were collected at shorter time intervals. Methyl orange showed rapid adsorption, with almost 90% removal during the first 10 min at both LDH dosages, corresponding to adsorption rates of 0.72 mg dye g⁻¹ LDH min⁻¹ and 1.41 mg dye g⁻¹ LDH min⁻¹, respectively. As expected (Zubair et al., 2017), higher LDH dosages resulted in greater dye removal efficiencies within shorter periods, particularly during the first 5 min. After 20 min, apparent removal efficiencies of 99% and 96% were achieved using 2.5 and 1.25 mg mL⁻¹ LDH, respectively. These results suggest that, at an LDH dosage of 5 mg mL⁻¹, 99% color removal was probably achieved before the first sample was collected at 30 min.

In contrast, crystal violet exhibited slower adsorption rates, ranging from 0.03 mg dye g⁻¹ LDH min⁻¹ at the highest adsorbent concentration of 5 mg mL⁻¹

to 0.073 mg dye g⁻¹ LDH min⁻¹ at the lowest adsorbent dose of 1.25 mg mL⁻¹ during the first 20 min. Consequently, only 30% color removal was achieved after 20 min using 5 mg mL⁻¹ LDH, and almost complete decolorization was not observed until 12 h. Again, decreasing the adsorbent dosage resulted in less efficient dye removal. After 12 h, the highest LDH dosage of 5 mg mL⁻¹ achieved up to 99% crystal violet removal, whereas the lowest dosage of 1.25 mg mL⁻¹ achieved 87% removal.

The stronger affinity observed for methyl orange can be related to the positively charged layers of Mg–Al LDHs, which may favor electrostatic attraction and anion exchange with the anionic dye (Yadav & Dasgupta, 2022; Zubair et al., 2017; Lu et al., 2024). Hydrogen bonding involving surface hydroxyl groups may also contribute to adsorption (Zubair et al., 2017). In contrast, the cationic nature of crystal violet results in less favorable electrostatic interactions, consistent with its slower uptake, although hydrogen bonding and interactions with accessible surface sites may still occur (Hasan et al., 2021). These mechanisms should be regarded as plausible interpretations rather than processes directly demonstrated in this study, since the relative contribution of electrostatic, ion-exchange, and other surface interactions cannot be quantified from the available data.

Taken together, these results indicate that the contrasting adsorption behavior of methyl orange and crystal violet cannot be explained by the surface area of the LDH alone. Previous studies using ligand-functionalized adsorbents have shown that selective uptake can be strongly influenced by the chemical functionality and affinity of specific binding

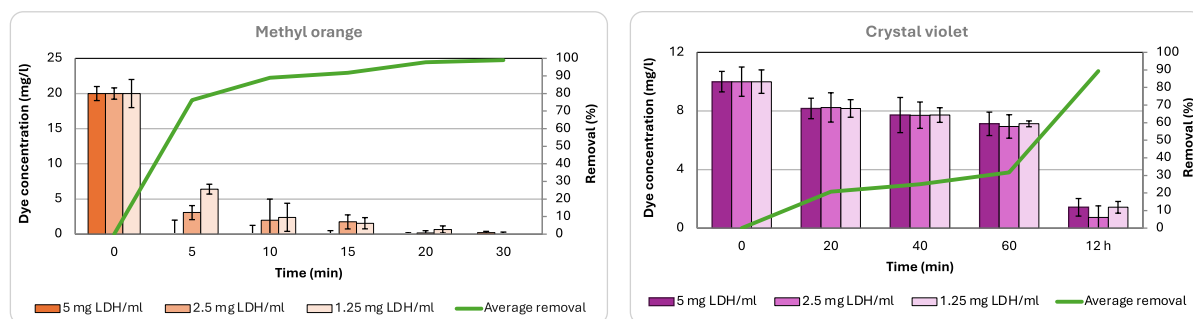


Fig. 2 Effect of contact time and LDH dosage on dye concentration and decolorization efficiency in the individual adsorption experiments. Error bars represent the standard deviation of replicate measurements

sites (Shahat et al., 2021; Awual et al., 2019; Awual et al., 2017). In the present system, the different ionic characteristics of the two dyes and their compatibility with the LDH surface also appear to influence adsorption. Furthermore, increasing the LDH dosage increased the number of available adsorption sites and consequently improved the overall removal efficiency. However, the higher mass-normalized adsorption rates observed at lower LDH dosages suggest that a greater proportion of the available sites was occupied.

3.2.2 Adsorption Kinetic Analysis

The time-dependent adsorption data were further evaluated using PFO, PSO, and intraparticle-diffusion models (Table S2). For methyl orange, both PFO and PSO equations provided a good description of the rapid adsorption process. At an LDH dosage of 1.25 mg mL^{-1} , the PFO model provided a slightly better fit ($R^2 = 0.9994$, $\text{RMSE} = 0.147 \text{ mg g}^{-1}$) than the PSO model ($R^2 = 0.9982$, $\text{RMSE} = 0.249 \text{ mg g}^{-1}$). The fitted PFO adsorption capacity (15.72 mg g^{-1}) was also close to the experimental value reached after 20 min (15.68 mg g^{-1}). At 2.5 mg mL^{-1} LDH, the PSO model showed the lowest RMSE and a slightly higher R^2 (0.9971) than the PFO model (0.9944), with a fitted q_e of 8.11 mg g^{-1} .

The intraparticle-diffusion model provided poorer fits for methyl orange ($R^2 = 0.9531$ and 0.8669 for 1.25 and 2.5 mg mL^{-1} LDH, respectively), and the fitted lines exhibited non-zero intercepts. These results suggest that intraparticle diffusion was not the sole process controlling the rapid methyl orange uptake. Nevertheless, the similarly high performance of both PFO and PSO models indicates that kinetic-model fitting alone is insufficient to assign a specific adsorption mechanism.

Crystal violet exhibited slower uptake. During the first 60 min, the intraparticle-diffusion model showed slightly higher R^2 values (0.9935 and 0.9944) than the PFO and PSO models, with intercepts close to zero. This behavior is consistent with a greater contribution of diffusion-related transport during the initial adsorption period. However, only four time points were available in this interval, and therefore a unique rate-controlling mechanism cannot be established from these data. Moreover, crystal violet had not reached equilibrium after 60 min. Removal increased from approximately 30–35% after 60 min to 87–94% after 12 h. Because the 12-h measurement

was obtained after a sedimentation period, it was not included in the kinetic fitting. Consequently, the fitted q_e values obtained for crystal violet should be interpreted only as apparent parameters describing the initial 0–60 min adsorption period and not as equilibrium adsorption capacities.

Overall, the kinetic analysis quantitatively confirms the markedly different adsorption behavior of the two dyes: methyl orange rapidly approached a plateau within the first 20 min, whereas crystal violet adsorption proceeded much more slowly. This difference is consistent with the more favorable interaction of the anionic methyl orange with the positively charged LDH layers, whereas adsorption of the cationic crystal violet is less electrostatically favored.

3.2.3 Dyes Decolorization through UV Irradiation

The color removal of methyl orange and crystal violet by UV irradiation is shown in Fig. 3. Decolorization was slow for both dyes. After 1 h, only 11% of methyl orange was removed, while crystal violet removal was slightly higher, reaching 22%. In fact, the pseudo-first-order rate constants for the decolorization of methyl orange and crystal violet were relatively low, at 0.0027 and 0.0045 min^{-1} , respectively (Malik & Sanyal, 2004). Consequently, substantial removal of both dyes, exceeding 60%, was not achieved until 12 h of exposure time to UV light.

The observed removal of dyes under UV irradiation may be associated with the formation of hydroxyl radicals through the photolysis of water. However, water photolysis is not highly efficient at 254 nm, limiting dye removal. Accordingly, low methyl orange removal under UV irradiation was also reported by (Haji et al., 2011), who obtained 26% removal after 4 h, and by (Navarro et al., 2019), who observed no removal after 90 s. Similarly, (Alshamsi et al., 2007) reported no crystal violet removal after 30 min of UV treatment. In the presence of external oxidizing agents such as H_2O_2 , the formation of hydroxyl radicals is enhanced, and faster removal rates can therefore be achieved for these and other dyes (Haji et al., 2011; Navarro et al., 2019; Alshamsi et al., 2007; Collivignarelli et al., 2019).

3.2.4 Dyes Oxidation through H_2O_2

Chemical oxidation with H_2O_2 is widely used for textile wastewater treatment because of its relatively

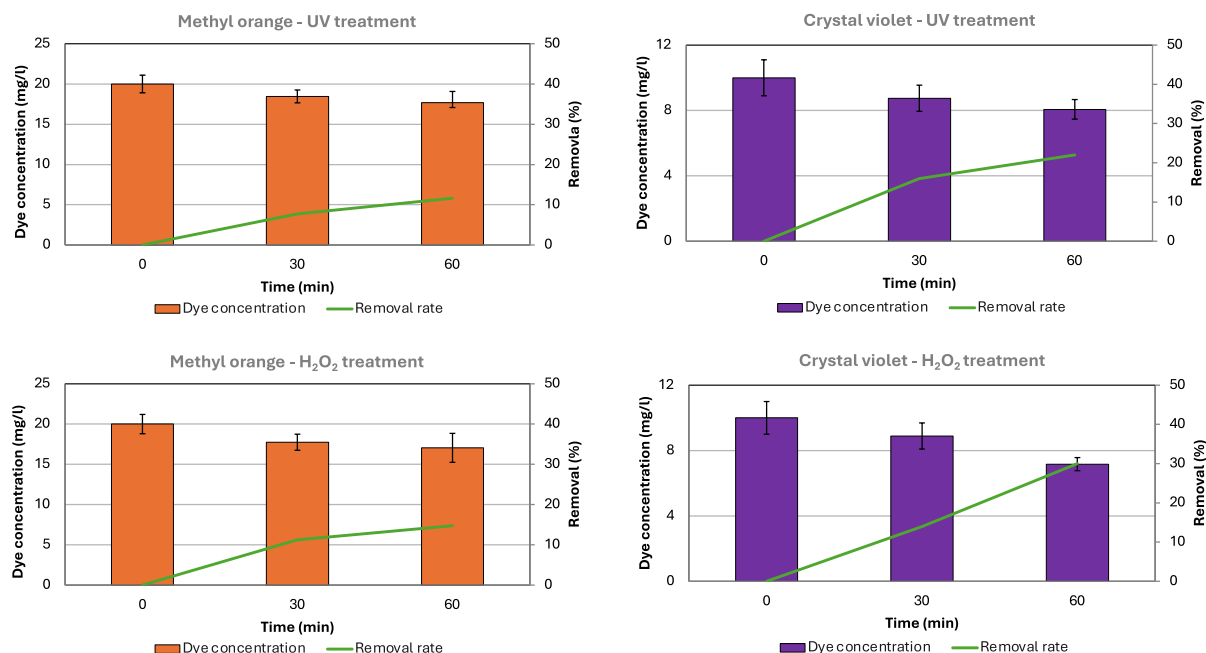


Fig. 3 Evolution of dye concentration and color removal efficiency as a function of treatment time during the individual UV irradiation and H_2O_2 oxidation experiments. Error bars represent the standard deviation of replicate measurements

simple application. Its main limitation is that the oxidation process usually requires activation, for example by UV light, to be effective (Ferreira et al., 2007). Consistent with this limitation, Fig. 3 shows that chemical oxidation with H_2O_2 alone resulted in a low decolorization efficiency for methyl orange, with 14% color removal after 1 h, while crystal violet color removal reached 30% after 1 h. Other studies found no decolorization of methyl orange (Navarro et al., 2019) or crystal violet (Foroutan et al., 2021), after treatment times of 90 s and 30 min, respectively.

The responses obtained for both dyes were similar to those observed under the UV irradiation alone, although the decolorization efficiencies and the pseudo-first-order rate constants were slightly higher in the presence of H_2O_2 . The rate constants were 0.0040 min^{-1} for methyl orange and 0.0055 min^{-1} for crystal violet. Taken together, these preliminary results suggest that crystal violet may be more susceptible than methyl orange to both UV irradiation and H_2O_2 treatment and, potentially, to their combined application.

The limited performance of H_2O_2 in the absence of irradiation is consistent with the need to activate the oxidant to promote the formation of highly reactive species (Malik & Sanyal, 2004). Under UV

irradiation, H_2O_2 photolysis can generate hydroxyl radicals and consequently enhance the attack on dye chromophores (Behera et al., 2021). However, the present study did not include radical-quenching experiments or EPR/ESR measurements. Therefore, hydroxyl-radical formation is proposed as a mechanism consistent with established UV/ H_2O_2 chemistry, but its contribution under the specific experimental conditions used here was not directly demonstrated.

3.3 Combined Effect of LDH, UV Irradiation and H_2O_2 on Color Removal

3.3.1 Results of the Box-Behnken Experimental Design

To evaluate the joint effects of the three variables on color removal, an incomplete 3^3 factorial design was employed, resulting in 15 experiments for each dye (Table 4). Experiments 13–15 were replicates of the center point and were used to assess experimental error. The results varied considerably, with decolorization efficiencies ranging from 2 to 98% for methyl orange and from 14 to 99% for crystal violet. The center-point replicates showed low

Table 4 Experimental conditions expressed as coded independent variables and corresponding decolorization responses for the dependent variables

Experiment number	INDEPENDENT			DEPENDENT	
	x_1	x_2	x_3	y_1	y_2
1	0	-1	-1	95	28
2	0	1	-1	98	87
3	0	-1	1	60	92
4	0	1	1	94	84
5	-1	-1	0	2	14
6	-1	1	0	76	90
7	1	-1	0	98	95
8	1	1	0	97	99
9	-1	0	-1	6	16
10	-1	0	1	85	78
11	1	0	-1	96	88
12	1	0	1	90	97
13	0	0	0	90	99
14	0	0	0	91	99
15	0	0	0	84	99

x_1 (LDH concentration; mg mL⁻¹), x_2 (UV irradiation time; h), x_3 (H₂O₂ dose; % v/v), y_1 (methyl orange removal; %), y_2 (crystal violet removal; %)

variability, with coefficients of variation below 10% in all cases. Thus, the design and the resulting analysis enabled a systematic evaluation of the variables affecting color-removal efficiency and indicated satisfactory experimental reproducibility.

Experiments 5, in which 5% H₂O₂ was applied without LDH or UV irradiation, and experiment 9, in which UV irradiation was applied alone, produced the lowest decolorization efficiencies. As observed in the preliminary experiments, hydrogen peroxide and UV irradiation applied individually were insufficient to achieve effective decolorization. According to the literature, this could be explained by the limited generation of reactive species when

the treatments were applied separately (Haji et al., 2011; Alshamsi et al., 2007).

The absence of H₂O₂ and UV irradiation in experiment 1, which used 2.5 mg mL⁻¹ LDH for methyl orange and 5 mg mL⁻¹ LDH for crystal violet, highlighted the effectiveness of LDH for the apparent removal of methyl orange, with an efficiency above 90%, but not of crystal violet, for which removal was only 28% after 10 min. As previously discussed, this selectivity may be attributed to the differing ionic characteristics of the two dyes, which result in different electrostatic interactions with the LDH surface. The highest decolorization efficiencies for both dyes were obtained in experiment 8, which used 5 mg mL⁻¹ LDH for methyl orange or 10 mg mL⁻¹ LDH for crystal violet, 2 h of UV irradiation, and 5% H₂O₂. This result is consistent with an additional contribution of UV/H₂O₂-induced decolorization to the overall response when combined with LDH adsorption. However, the specific reactive species involved were not identified or quantified in the present study.

3.3.2 Determination and Statistical Assessment of the Regression Models

Multiple regression analysis was applied to the experimental design matrix and the color removal responses shown in Table 4. The regression coefficients, standard errors, t-values, p-values, and 95% confidence intervals are provided in Table S3. The predicted-versus-observed plot is shown in Figure S2 and Table S4. Regression terms with *p*-values above 0.05 were removed where appropriate, while lower-order terms were retained when required to preserve model hierarchy. The resulting reduced empirical polynomial equations, expressed in coded form, were used to model the color removal efficiencies of methyl orange and crystal violet (Eqs. 9 and 10):

$$MO(\%) = 88.3 + 23.8x_1 - 18.8x_1^2 + 9.3x_2 - 9.8x_3 - 18.8x_1x_2 - 21.3x_1x_3 + 28.0x_1^2x_3 \quad (9)$$

$$CV(\%) = 99.0 + 22.8x_1 - 13.9x_1^2 + 12.5x_2 - 10.6x_2^2 + 15x_3 - 15.4x_3^2 - 18x_1x_2 + 7.5x_1^2x_2 - 13.3x_1x_3 - 16.5x_2x_3 \quad (10)$$

where x_1 , x_2 , and x_3 are the dimensionless coded variables corresponding to LDH dosage, UV irradiation time, and H_2O_2 stock-solution dosage, respectively, as defined in Eqs. (2) and (3).

Based on the coefficients of determination obtained from the original regression analyses, the models were globally statistically significant. The methyl orange model showed $R^2 = 0.9630$ and adjusted $R^2 = 0.9261$. The crystal violet model showed $R^2 = 0.9987$ and adjusted $R^2 = 0.9954$. The agreement between the predicted and experimental responses for the 15 Box–Behnken runs is presented in Figure S2 and Table S4. To further assess predictive performance, additional validation experiments were conducted using combinations of the independent variables within the experimental domain, as presented in Table 5. For methyl orange, the experimentally observed and model-predicted decolorization efficiencies were 44% and 57%, respectively, indicating appreciable uncertainty in the individual prediction. In contrast, the experimental decolorization efficiency for crystal violet was approximately 99%, in agreement with the near-complete decolorization predicted by the model.

3.3.3 Effects of Model Components and their Interactions

The Pareto charts (Figure S3) show the relative magnitude of the standardized effects, while their direction was determined from the regression coefficients in Eqs. (9) and (10). For methyl orange, LDH dosage was the dominant positive factor, confirming the major contribution of adsorption to its removal. UV irradiation had a smaller positive effect, whereas H_2O_2 showed a negative linear effect within the investigated domain. The interaction terms further indicate that the treatment components acted in a condition-dependent and non-additive manner. For

crystal violet, LDH dosage also exerted the strongest positive effect, but UV irradiation and H_2O_2 both made substantial positive contributions. The negative interaction and quadratic terms indicate response curvature and diminishing benefits at higher treatment levels. Overall, UV/ H_2O_2 contributed more markedly to crystal violet decolorization than to methyl orange removal. However, the statistical analysis does not allow the response to be partitioned into exact contributions from adsorption, photolysis, and oxidation, nor does it demonstrate chemical synergy between Mg–Al LDH and H_2O_2 /UV.

3.3.4 Three-Dimensional (3D) Response Surfaces

To better understand the effects of the three treatments and their interactions on the dye decolorization, three-dimensional response surfaces were constructed using the regression model equations. Since the models contained three independent variables, the response surfaces were expressed as a function of two variables, namely LDH dosage and UV irradiation time, while the H_2O_2 dosage was maintained at 0%, 5%, or 10% (v/v), corresponding to coded values of -1 , 0 , and $+1$, respectively. Three response-surface plots were therefore obtained for each dye (Fig. 4). The curvature of the response surfaces and the significance of several interaction terms showed that the experimental responses could not be adequately represented by simple linear models. The final equations were therefore expressed as hierarchical reduced polynomial models containing quadratic and selected mixed third-order terms. Accordingly, these equations should be interpreted as empirical models applicable within the investigated design space rather than as conventional quadratic response-surface models.

Figure 4a shows the effects of the most influential operational variables, LDH dosage and UV irradiation time, at the lowest H_2O_2 dosages. For methyl

Table 5 Validation experiments comparing the experimental decolorization efficiencies with the values predicted by the equations derived from the Box–Behnken design

Dye tested	Dependent variables				Independent variable Dye removal (%)	
	Value	x_1	x_2	x_3	Experimental	Predicted
Methyl orange	Coded	-0.5	-1	0.5	44	57
	Real	1.25 mg LDH mL ⁻¹	0 h UV	7.5% H_2O_2		
Crystal violet	Coded	0.5	1	-0.5	99	100
	Real	7.5 mg LDH mL ⁻¹	1 h UV	2.5% H_2O_2		

orange, the model predicted higher color removal when the LDH concentration increases, although the maximum value was obtained at approximately 4 mg mL^{-1} of LDH, corresponding to a coded value of 0.6. The effect of UV irradiation was only slightly relevant under low LDH dosages. In contrast, crystal violet removal increased substantially with longer UV exposure times and higher LDH dosages (Fig. 4a). In this case, the response surface highlights that the combined application of LDH adsorption and UV irradiation improved crystal violet decolorization. At an LDH dosage of approximately 4 mg mL^{-1} and UV irradiation times longer than 50 min, decolorization efficiencies greater than 80% were predicted.

The response surfaces obtained at an H_2O_2 dosage of 5% (v/v) (Fig. 4b) were similar to those obtained in the absence of the oxidant. For methyl orange, the addition of H_2O_2 favored decolorization mainly when no LDH was added but UV irradiation was applied, suggesting the dye is been oxidized by a UV/ H_2O_2 advanced oxidation process. Similarly, crystal violet decolorization under UV irradiation increased in the presence of H_2O_2 . The oxidant also appeared to enhance the overall apparent removal obtained in the presence of LDH. In fact, near-complete crystal violet decolorization was achieved using 4 mg mL^{-1} LDH and 50 min of UV irradiation. However, when the H_2O_2 dosage was increased to 10% (Fig. 4c) the predicted removal of crystal violet decreased to approximately 90%. The response surfaces obtained at 10% H_2O_2 indicated that the removal of both dyes is impaired at the highest oxidant dosage (Fig. 4c).

Overall, nonlinear relationships and significant interactions among the treatment variables were observed, as reflected in the response surfaces. Thus, the effects of the treatments under combined conditions could not be inferred directly from their individual effects, highlighting the need to optimize their operating conditions jointly.

3.3.5 Optimal Treatment Combination for Maximizing Dye Removal Efficiency

The regression models were optimized using the Generalized Reduced Gradient nonlinear solving method to identify the conditions that maximized color removal efficiency. The results are summarized in Table 6. For methyl orange, the combination of 3.2 mg mL^{-1} LDH and 2 h of UV irradiation, without

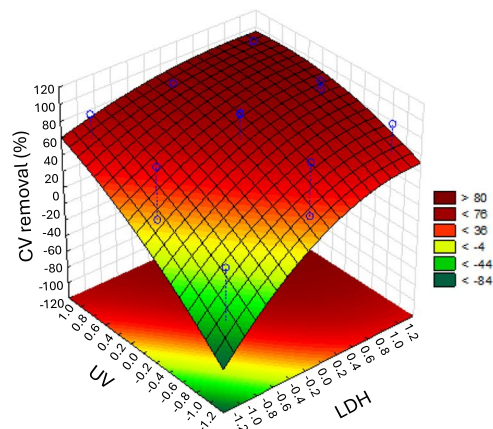
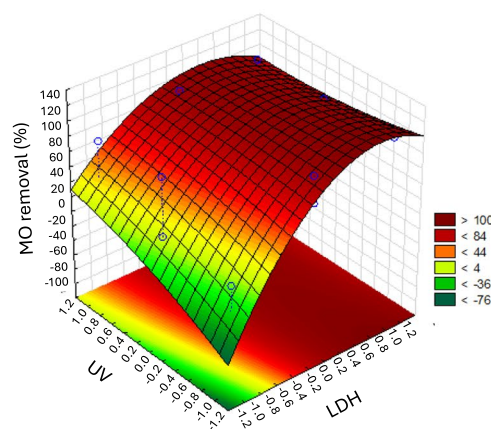
H_2O_2 , provided the numerical solution that maximized the predicted response. However, almost the same removal efficiency could be achieved by eliminating UV irradiation and increasing the LDH dosage to 4.2 mg mL^{-1} . This result supports the dominant contribution of LDH adsorption to methyl orange removal and indicates that UV irradiation provided only a limited additional benefit near the predicted maximum.

In contrast, maximizing crystal violet removal required the combined application of the three treatments. The predicted optimum consisted of 8.7 mg mL^{-1} LDH, 32 min of UV irradiation, and 5.7% H_2O_2 stock solution (Table 6). Additional experiments were conducted to validate the optimal conditions identified for both dyes, and removal efficiencies above 99% were obtained, as predicted.

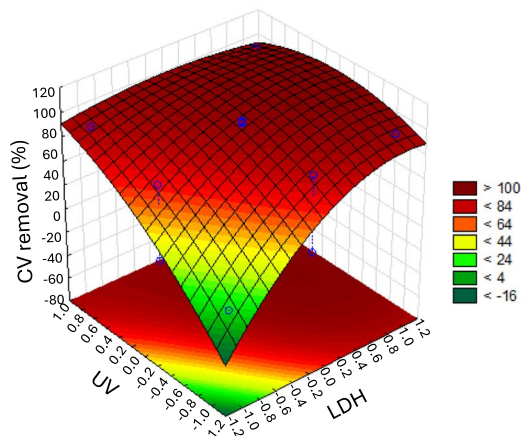
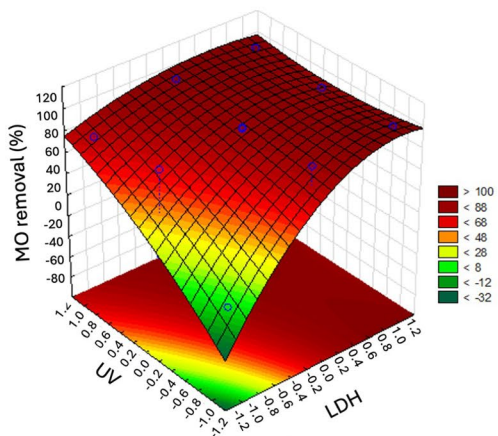
In addition to these absolute maxima, multiple local optima can be identified when an additional constraint is introduced to establish a specified removal efficiency. This means that different treatment combinations can be selected depending on treatment availability, cost, or other operational considerations while still achieving high removal efficiencies. For instance, if a textile facility uses both dyes, a treatment combining 4.5 mg mL^{-1} LDH, 1 h of UV irradiation, and a 3.5% dosage of the H_2O_2 stock solution could provide decolorization efficiencies greater than 98% for both dyes.

However, these optimized solutions were derived from controlled model-dye systems and do not reproduce the complexity of real textile wastewater. Matrix constituents such as salts, surfactants, auxiliary chemicals, organic matter, and dye mixtures may affect adsorption through competition and altered ionic interactions, while also reducing UV/ H_2O_2 efficiency through radical scavenging or UV attenuation. Therefore, the optimized conditions reported should be regarded as model-system optima and should not be directly extrapolated to real textile effluents. Future studies should evaluate the process using increasingly complex synthetic matrices and, ultimately, real textile wastewater.

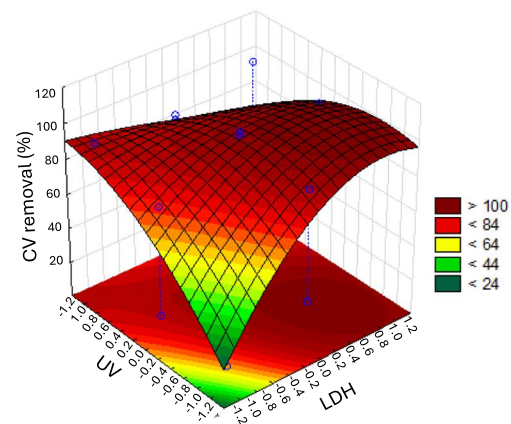
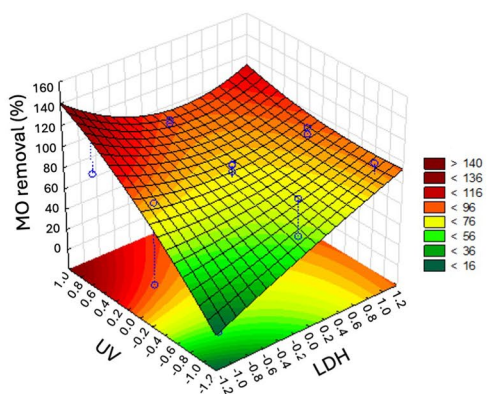
From a practical perspective, the common operating condition identified for both dyes requires a relatively high LDH dosage (4.5 kg LDH per cubic meter of treated water), 1 h of UV irradiation, and 3.5% v/v of the H_2O_2 stock solution. These requirements may increase chemical consumption, energy demand, solid handling, and residual oxidant management.



(a)



(b)



(c)

Fig. 4 3D response surfaces showing the predicted dependence of apparent dye removal (y) on LDH dosage (x_1) and UV irradiation time (x_2) at fixed H_2O_2 concentrations (a) 0%, (b) 5%, and (c) 10%. MO: methyl orange; CV: crystal violet

However, the recovery and reuse of the LDH over consecutive adsorption cycles, as discussed in the following section, could reduce fresh adsorbent consumption and sludge generation. Since specific UV energy consumption, residual H_2O_2 , spent-solid production, and the economic benefits of LDH reuse were not quantified, a reliable cost or feasibility assessment could not be performed. Therefore, this condition should be regarded as a model-system proof of concept rather than a full-scale treatment recommendation, and future studies should re-optimize the process for each wastewater matrix using both performance and resource-consumption criteria.

3.4 Reusability and Thermal Regeneration of the LDH

LDH reuse and regeneration after adsorption were evaluated to assess the performance of the adsorbent under repeated use. The experiment was conducted with methyl orange because this dye exhibited the highest and fastest adsorption onto the Mg–Al LDH, making it the most appropriate system for an initial assessment of adsorbent reuse. Accordingly, the results presented here refer specifically to methyl orange adsorption and should not be extrapolated to crystal violet, mixed-dye solutions, or the combined LDH/UV/ H_2O_2 treatment. The LDH maintained methyl orange removal efficiencies above 90% throughout ten consecutive adsorption cycles (Fig. 5), with values remaining between 98 and 99% during the first seven cycles. A gradual decrease was subsequently observed, although removal remained close to 90% after the tenth cycle. These results indicate promising short-term reusability of the LDH for methyl orange adsorption under the tested conditions.

After the tenth cycle, the adsorbent was thermally regenerated once at 600 °C for 30 min. The regenerated LDH achieved a methyl orange removal efficiency of 94%, indicating that its adsorption performance was partially restored after thermal treatment. Previous studies on methyl orange removal evaluated LDH reusability using different materials and regeneration protocols. A PEG-400-modified Mg–Al LDH showed a decrease in MO desorption efficiency from 93.64% to 88.23% over three regeneration cycles (Kameliya et al., 2024). A nitrate-intercalated Mg–Al LDH maintained nearly constant adsorption performance during the first three of five adsorption–regeneration cycles, followed by a decline in the fourth and fifth cycles (Yadav & Dasgupta, 2022). A more pronounced loss was reported for Mg₄Fe-LDH, whose MO removal efficiency decreased from 94.86% to 62.20% over four cycles (Lu et al., 2024). In comparison, the calcined Mg–Al LDH investigated here maintained removal efficiencies above 90% over ten consecutive adsorption cycles and achieved 94% removal after subsequent thermal regeneration, although differences in LDH composition and regeneration protocols prevent direct comparison.

In the present study, regeneration was performed only once, after the tenth adsorption cycle; therefore, the effect of repeated thermal regeneration on energy consumption and long-term adsorbent performance was not assessed, although regeneration at 600 °C may be energy-intensive and increase operating costs. In addition, LDH mass loss and post-use structural changes were not evaluated; therefore, its stability after repeated adsorption and thermal regeneration could not be determined. To complete this initial reusability and regeneration study, further work should therefore evaluate reuse with crystal violet, real textile wastewater, and the combined LDH/UV/ H_2O_2 treatment, together with adsorbent mass recovery, repeated regeneration cycles, energy requirements, and structural characterization before and after reuse. Nevertheless, the

Table 6 Optima values of the independent variables to maximize color removal

Dye tested	Dependent variables			
	Value	x_1	x_2	x_3
Methyl orange	Coded	0.28	1	−1
	Real	3.20 mg mL ^{−1} LDH	2 h UV	0% H_2O_2
Crystal violet	Coded	0.74	0.04	0.15
	Real	8.70 mg mL ^{−1} LDH	32 min UV	5.7% H_2O_2

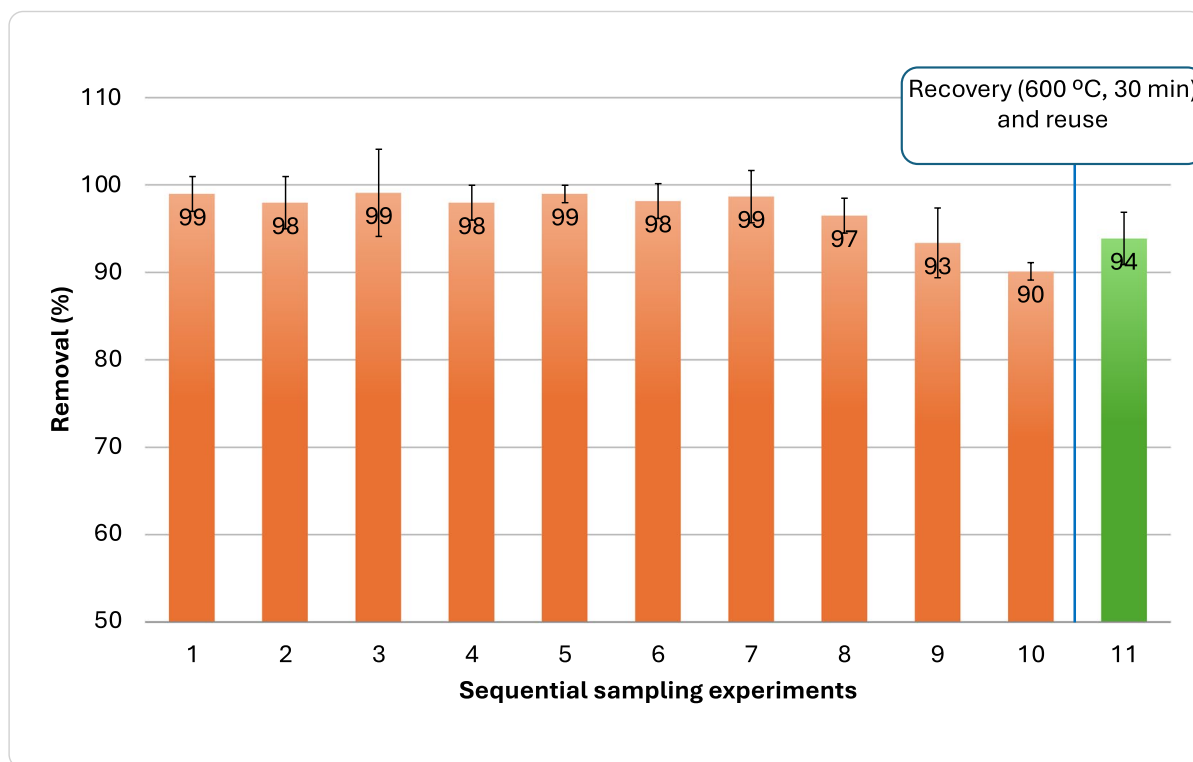


Fig. 5 Methyl orange removal efficiency of Mg–Al LDH over ten consecutive adsorption cycles and after thermal regeneration at 600 °C for 30 min and reuse in cycle 11. Bars represent

mean values, and error bars indicate the standard deviation of replicate measurements

maintenance of methyl orange removal above 90% over ten consecutive cycles and the achievement of 94% after thermal regeneration provide encouraging evidence of the reusability and regeneration potential of the synthesized Mg–Al LDH.

4 Conclusions

This study systematically compared LDH adsorption, UV irradiation, and H_2O_2 treatment, applied individually and in combination, for the decolorization and apparent aqueous-phase removal of two dyes with contrasting ionic characteristics. LDH adsorption was the dominant treatment for methyl orange, whereas crystal violet exhibited slower adsorption and a greater response to UV/ H_2O_2 , making the combined treatment more suitable for its rapid and near-complete decolorization.

For methyl orange, the optimal condition consisted of 3.2 mg mL⁻¹ LDH and 2 h of UV irradiation

without H_2O_2 , while a comparable removal efficiency could be achieved using 4.2 mg mL⁻¹ LDH alone. In contrast, near-complete crystal violet removal required 8.70 mg mL⁻¹ LDH, 32 min of UV irradiation, and 5.7% H_2O_2 stock solution. A generalized condition of 4.5 mg mL⁻¹ LDH, 1 h of UV irradiation, and 3.5% H_2O_2 stock solution achieved decolorization efficiencies above 98% for both dyes tested separately. These results highlight the importance of adapting the treatment combination to the ionic characteristics of the target dye.

The Mg–Al LDH also maintained methyl orange removal efficiencies above 90% over ten consecutive adsorption cycles, while a single thermal regeneration step at 600 °C partially restored the removal efficiency to 94%. These findings provide encouraging preliminary evidence of LDH reusability although they cannot be extrapolated directly to other dyes or wastewaters.

Although the results demonstrate effective decolorization and aqueous-phase dye removal, further TOC or COD, transformation-product, and toxicity analyses

are required to assess mineralization and environmental safety. Validation using real textile wastewater and evaluation of chemical and energy requirements are also necessary before scale-up. Overall, the LDH/UV/H₂O₂ system represents a flexible proof-of-concept approach for apparent removal of two model dyes with different ionic characteristics.

Acknowledgements The authors thank the Defense University Center at the Spanish Naval Academy (CUD-ENM) for all the support provided for this research (Project PICUD-2022-02). Authors would also like to acknowledge the use of scientific and technical services of Centro de Apoio Científico e Tecnológico á Investigación (CACTI-Universidade de Vigo). This work was supported by the D-ECO-FASTECH project (20261RO01KA220VET000449933), funded by the Erasmus+ Programme of the European Union. Funding for open access charge: Universidade da Coruña/CISUG.

Author Contributions R.D. and P.H. performed the experiments, analyzed the data, and prepared the first draft of the manuscript. E.A. conceived and supervised the study, contributed to the experimental design and data interpretation, and revised the manuscript. L.G. contributed to data interpretation, manuscript revision, conceived and supervised the study, contributed to the experimental design and data interpretation, revised the manuscript, and overall scientific discussion. All authors reviewed and approved the final manuscript.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Alshamsi, F. A., Albadwawi, A. S., Alnuaimi, M. M., Rauf, M. A., & Ashraf, S. S. (2007). Comparative efficiencies of the degradation of crystal violet using UV/hydrogen peroxide and Fenton's reagent. *Dyes and Pigments*, 74(2), 283–287. <https://doi.org/10.1016/j.dyepig.2006.02.016>
- Asamoah, E. N., Liu, H., & Fan, X. (2025). Fixed-bed adsorption of methylene blue using granular NaX zeolite/attapulgite composite: Efficiency, mechanism and reusability of saturated G-NaX/A. *Separation and Purification Technology*, 354, 128710. <https://doi.org/10.1016/j.seppur.2024.128710>
- Awual, M. R., Alharthi, N. H., Okamoto, Y., Karim, M. R., Halim, M. E., Hasan, M. M., Rahman, M. M., Islam, M. M., Khaleque, M. A., & Sheikh, M. C. (2017). Ligand field effect for Dysprosium(III) and Lutetium(III) adsorption and EXAFS coordination with novel composite nanomaterials. *Chemical Engineering Journal*, 320, 427–435. <https://doi.org/10.1016/j.cej.2017.03.075>
- Awual, M. R., Hasan, M. M., Islam, A., Rahman, M. M., Asiri, A. M., Khaleque, M. A., & Sheikh, M. C. (2019). Introducing an amine functionalized novel conjugate material for toxic nitrite detection and adsorption from wastewater. *Journal of Cleaner Production*, 228, 778–785. <https://doi.org/10.1016/j.jclepro.2019.04.280>
- Awual, M. R., Hasan, M. N., Hasan, M. M., Salman, M. S., Sheikh, M. C., Kubra, K. T., Islam, M. S., Marwani, H. M., Islam, A., Khaleque, M. A., Waliullah, R. M., Hossain, M. S., Rasee, A. I., Rehan, A. I., & Awual, M. E. (2023). Green and robust adsorption and recovery of Europium(III) with a mechanism using hybrid donor conjugate materials. *Separation and Purification Technology*, 319, 124088. <https://doi.org/10.1016/j.seppur.2023.124088>
- Bai, Z., Rong, D., Li, M., Xu, G., Liu, S., Zeng, J., Lv, Y., Tang, Y., & Wen, X. (2024). Hierarchical Mg/Al hydroxide hollow microspheres with excellent adsorption capability towards Congo red. *Dalton Transactions*, 53(8), 3744–3755. <https://doi.org/10.1039/d3dt03816e>
- Behera, M., Nayak, J., Banerjee, S., Chakraborty, S., & Tripathy, S. K. (2021). A review on the treatment of textile industry waste effluents towards the development of efficient mitigation strategy: An integrated system design approach. *Journal of Environmental Chemical Engineering*, 9(4), Article 105277. <https://doi.org/10.1016/j.jece.2021.105277>
- Bora, N., Joshi, D. P., & Sharma, R. (2026). Contamination toxicity and remediation of stable dyes and oils in aquatic wastewater systems. *Discover Environment*, 4(1), 166. <https://doi.org/10.1007/s44274-026-00680-w>
- Bour, A., Budde Christensen, T., Hunka, A. D., Palmqvist, A., Skjold, E., & Syberg, K. (2023). Implications of circular textile policies for the future regulation of hazardous substances in textiles in the European Union. *Science of the Total Environment*, 896, Article 165153. <https://doi.org/10.1016/j.scitotenv.2023.165153>
- Box, G.E.P., & Behnken, D.W. (1960). Simplex-sum designs—A class of 2nd order rotatable designs derivable of those of 1st order. *The Annals of Mathematical Statistics*, 31, 838–864.

- Collivignarelli, M. C., Abbà, A., Carnevale, Miino M., & Damiani, S. (2019). Treatments for color removal from wastewater: State of the art. *Journal of Environmental Management*, 236, 727–745. <https://doi.org/10.1016/j.jenvman.2018.11.094>
- Devesa-Rey, R., Del Val, J., Feijoo, J., González-Coma, J., Castiñeira, G., & González-Gil, L. (2021). Preparation of synthetic clays to remove phosphates and ibuprofen in water. *Water (Switzerland)*, 13(17), 2394. <https://doi.org/10.3390/w13172394>
- Díaz, O., González, E., Rodríguez-Sevilla, J., & Vera, L. (2024). Nutrient removal from secondary wastewater effluent by adsorption with hydrotalcite: Effect of calcination temperature and co-existing ions. *Water Science and Technology*, 90(6), 1814–1833. <https://doi.org/10.2166/wst.2024.304>
- Ferreira, S. L. C., Bruns, R. E., Ferreira, H. S., Matos, G. D., David, J. M., Brandão, G. C., da Silva, E. G. P., Portugal, L. A., dos Reis, P. S., Souza, A. S., & dos Santos, W. N. L. (2007). Box-Behnken design: An alternative for the optimization of analytical methods. *Analytica Chimica Acta*, 597(2), 179–186. <https://doi.org/10.1016/j.aca.2007.07.011>
- Foroutan, R., Peighambari, S. J., Peighambari, S. H., Pateiro, M., & Lorenzo, J. M. (2021). Adsorption of crystal violet dye using activated carbon of lemon wood and activated carbon/Fe₃O₄ magnetic nanocomposite from aqueous solutions: A kinetic, equilibrium and thermodynamic study. *Molecules*, 26(8), 2241. <https://doi.org/10.3390/molecules26082241>
- Fripiat, J. J., Gatineau, L., & van Damme, H. (1986). Multi-layer physical adsorption on fractal surfaces. *Langmuir*, 2(5), 562–567. <https://doi.org/10.1021/la00071a006>
- Gupta, D., Singh, C., Koranne, A., Singh, S., Azad, U. P., Singh, A. K., & Singh, S. K. (2024). Layered Double Hydroxide: An Economical and Dynamic Choice for Water Remediation with Multiple Functionality. *ChemistrySelect*, 9(39), e202303300. <https://doi.org/10.1002/slct.202303300>
- Haji, S., Benstaali, B., & Al-Bastaki, N. (2011). Degradation of methyl orange by UV/H₂O₂ advanced oxidation process. *Chemical Engineering Journal*, 168(1), 134–139. <https://doi.org/10.1016/j.cej.2010.12.050>
- Hamzah, M. A. A. M., Yusof, N., Jaafar, J., Ismail, A. F., Lau, W. J., Salleh, W. N. W., & Aziz, F. (2025). Transforming palm oil fuel ash into high-performance biosorbents: One-pot synthesis of layered double hydroxide composites for efficient removal of hazardous dyes from textile wastewater. *Water, Air, and Soil Pollution*, 236(10), Article 610. <https://doi.org/10.1007/s11270-025-08268-5>
- Hasan, M. M., Shenashen, M. A., Hasan, M. N., Znad, H., Salman, M. S., & Awual, M. R. (2021). Natural biodegradable polymeric bioadsorbents for efficient cationic dye encapsulation from wastewater. *Journal of Molecular Liquids*, 323, 114587. <https://doi.org/10.1016/j.molliq.2020.114587>
- Hasan, M. M., Kubra, K. T., Hasan, M. N., Awual, M. E., Salman, M. S., Sheikh, M. C., Rehan, A. I., Rasee, A. I., Waliullah, R. M., Islam, M. S., Khandaker, S., Islam, A., Hossain, M. S., Alsukaibi, A. K. D., Alshammari, H. M., & Awual, M. R. (2023). Sustainable ligand-modified based composite material for the selective and effective cadmium(II) capturing from wastewater. *Journal of Molecular Liquids*, 371, Article 121125. <https://doi.org/10.1016/j.molliq.2022.121125>
- Hassanzadeh, A., Hamidpour, M., Dahaji, P. A., Akhgar, A., & Kariman, K. (2024). Maximizing the efficiency of layered double hydroxides as a slow-release phosphate fertilizer: A study on the impact of plant growth-promoting rhizobacteria. *Applied Clay Science*, 262, 107620. <https://doi.org/10.1016/j.clay.2024.107620>
- Kameliya, J., Leel, N. S., Mukeshchandra, G. S., Khalifa, M. E., El-Bahy, S. M., & Vyas, S. (2024). Rapid removal of methyl orange from aqueous media by modified Mg–Al layered double hydroxides. *Journal of the Indian Chemical Society*, 101(10), Article 101355. <https://doi.org/10.1016/j.jics.2024.101355>
- Khuong, D. A., Kieu, T. T., Nakaoka, Y., Tsubota, T., Tashima, D., Nguyen, H. N., & Tanaka, D. (2022). The investigation of activated carbon by K₂CO₃ activation: Micropores- and macropores-dominated structure. *Chemosphere*, 299, 134365. <https://doi.org/10.1016/j.chemosphere.2022.134365>
- Khushnood, S., Khan, J. R., Hussain, S., Khan, H., Wahab, F., Ng, K. C., & Shahzad, M. W. (2024). Synthesis and characterization of mesoporous caged silica for efficient adsorption of methylene blue from aqueous solution. *International Communications in Heat and Mass Transfer*, 156, 107706. <https://doi.org/10.1016/j.icheatmasstransfer.2024.107706>
- Kishor, R., Purchase, D., Saratale, G. D., Romanholo Ferreira, L. F., Hussain, C. M., Mulla, S. I., & Bharagava, R. N. (2021). Degradation mechanism and toxicity reduction of methyl orange dye by a newly isolated bacterium *Pseudomonas aeruginosa* MZ520730. *Journal of Water Process Engineering*, 43, 102300. <https://doi.org/10.1016/j.jwpe.2021.102300>
- Klaka, K., & Schlittmeier, S. J. (2026). Fast or fair fashion? Experimental evidence on psychological drivers of perceived sustainability and consumption intention. *Frontiers in Sustainability*, 7, Article art. no. 1736995. <https://doi.org/10.3389/frsus.2026.1736995>
- Kubra, K. T., Salman, M. S., Hasan, M. N., Islam, A., Teo, S. H., Hasan, M. M., Sheikh, M. C., & Awual, M. R. (2021). Sustainable detection and capturing of cerium(III) using ligand embedded solid-state conjugate adsorbent. *Journal of Molecular Liquids*, 338, 116667. <https://doi.org/10.1016/j.molliq.2021.116667>
- Li, X., Li, X., Chen, G., Li, H., Duan, Y., Sun, Y., Tiraferri, A., & Liu, B. (2025). Efficient recovery of lithium from spent lithium-ion battery raffinate by Mn and Al-based adsorbents: Pretreatment, adsorption mechanism, and performance comparison. *Separation and Purification Technology*, 354, 128652. <https://doi.org/10.1016/j.seppur.2024.128652>
- Lu, L., Bi, J., Shahab, A., Zhang, H., Rad, S., & Wu, Z. (2024). Optimizing Mg-Fe layered double hydroxides and efficient removal of methyl orange: Screening, performance and mechanism. *Environmental Technology and Innovation*, 34, 103629. <https://doi.org/10.1016/j.eti.2024.103629>
- Malik, P. K., & Sanyal, S. K. (2004). Kinetics of decolourisation of azo dyes in wastewater by UV/H₂O₂ process. *Separation and Purification Technology*, 36(3), 167–175. [https://doi.org/10.1016/S1383-5866\(03\)00212-0](https://doi.org/10.1016/S1383-5866(03)00212-0)
- Mani, S., & Bharagava, R. N. (2016). Exposure to crystal violet, its toxic, genotoxic and carcinogenic effects on environment and its degradation and detoxification for environmental safety. *Reviews of Environmental Contamination and Toxicology*, 237, 71–104. https://doi.org/10.1007/978-3-319-23573-8_4
- Masoud, A. M., Mubark, A. E., Taha, M. H., & Ibrahim, S. (2024). Nanostructured layered double hydroxide (NLDH) - Zn/Al-based materials: Strategy to improve performance

- for zirconium sorption from acidic sulfate solution. *RSC Advances*, 14(39), 28455–28468. <https://doi.org/10.1039/d4ra03845b>
- Meiramkulova, K., Kydyrbekova, A., Mkilima, T., Bazarbayeva, T., Sholpan, U., Zarbaliyev, M., Uryumtseva, T., Kurbanova, L., Tussupova, B., & Zhumagulov, M. (2024). Calcination-modified zeolite filtration and UV disinfection for contaminant Mitigation in groundwater. *Case Studies in Chemical and Environmental Engineering*, 10, 100781. <https://doi.org/10.1016/j.csee.2024.100781>
- Munonde, T. S., Madima, N., Ratshiedana, R., NosizoNomngongo, P., Mofokeng, L. E., & Dima, R. S. (2025). Synergistic adsorption-photocatalytic remediation of methylene blue dye from textile industry wastewater over NiFe LDH supported on tyre-ash derived activated carbon. *Applied Surface Science*, 679, 161205. <https://doi.org/10.1016/j.apsusc.2024.161205>
- Navarro, P., Pellicer, J. A., & Gómez-López, V. M. (2019). Degradation of azo dye by an UV/H₂O₂ advanced oxidation process using an amalgam lamp. *Water and Environment Journal*, 33(3), 476–483. <https://doi.org/10.1111/wej.12418>
- Pinto, V. C. D., & Peleg Mizrahi, M. (2025). The Health Impact of Fast Fashion: Exploring Toxic Chemicals in Clothing and Textiles. *Encyclopedia*, 5(2), 84. <https://doi.org/10.3390/encyclopedia5020084>
- Qi, R., Qian, C., Li, Y., & Wang, Y. (2024). Biofilm formation on MgFe-LDH@quartz sand as novel wetland substrate under varied C/N ratios for BDE-47 removal. *Environmental Pollution*, 361, 124779. <https://doi.org/10.1016/j.envpol.2024.124779>
- Rajabinejad, H., Bucişcanu, I.-I., & Maier, S.S. (2019). Current approaches for raw wool waste management and unconventional valorization: A review. *Environmental Engineering and Management Journal*, 18(7), 1439–1456. <https://doi.org/10.30638/eeemj.2019.136>
- Salman, M. S., Hasan, M. N., Hasan, M. M., Kubra, K. T., Sheikh, M. C., Rehan, A. I., Waliullah, R. M., Rasee, A. I., Awual, M. E., Hossain, M. S., Alsukaibi, A. K. D., Alshammari, H. M., & Awual, M. R. (2023). Improving copper(II) ion detection and adsorption from wastewater by the ligand-functionalized composite adsorbent. *Journal of Molecular Structure*, 1282, Article 135259. <https://doi.org/10.1016/j.molstruc.2023.135259>
- Senthil Rathil, B., & Senthil Kumar, P. (2022). Sustainable approach on the biodegradation of azo dyes: A short review. *Current Opinion in Green and Sustainable Chemistry*, 33, 100578. <https://doi.org/10.1016/j.cogsc.2021.100578>
- Shahat, A., Kubra, K. T., Salman, M. S., Hasan, M. N., & Hasan, M. M. (2021). Novel solid-state sensor material for efficient cadmium(II) detection and capturing from wastewater. *Microchemical Journal*, 164, 105967. <https://doi.org/10.1016/j.microc.2021.105967>
- Shu, H.-Y. (2006). Degradation of dyehouse effluent containing C.I. Direct Blue 199 by processes of ozonation, UV/H₂O₂ and in sequence of ozonation with UV/H₂O₂. *Journal of Hazardous Materials*, 133(1–3), 92–98. <https://doi.org/10.1016/j.jhazmat.2005.09.056>
- Stanton, T., Stanes, E., Gwinnett, C., Lei, X., Cauilan-Cureg, M., Ramos, M., Sallach, J. B., Harrison, E., Osborne, A., Sanders, C. H., Baynes, E., Law, A., Johnson, M., Ryves, D. B., Sheridan, K. J., Blackburn, R. S., & McKay, D. (2023). Shedding off-the-grid: The role of garment manufacturing and textile care in global microfibre pollution. *Journal of Cleaner Production*, 428, 139391. <https://doi.org/10.1016/j.jclepro.2023.139391>
- Su, J., Zhao, M., Zhao, Z., Sun, L., & Meng, Z. (2024). Mg-Al layered double hydroxides modified by 2,4,6-trimercaptos-triazine for effective removal heavy metals from soil. *Chemical Engineering Science*, 300, 120563. <https://doi.org/10.1016/j.ces.2024.120563>
- Teow, Y. H., Zulkifli, E., & Wikramasinghe, S. R. (2023). Performance and resilience of the PolyCera® Titan membrane for industrial wastewater treatment. *Water Science and Technology*, 87(5), 1056–1071. <https://doi.org/10.2166/wst.2023.034>
- Urbina-Suarez, N. A., Angel-Ospina, A. C., Lopez-Barrera, G. L., Barajas-Solano, A. F., & Machuca-Martínez, F. (2024). S-curve and landscape maps for the analysis of trends on industrial textile wastewater treatment. *Environmental Advances*, 15, Article 100491. <https://doi.org/10.1016/j.envadv.2024.100491>
- Usgodaarachchi, L., Thambiliyagodage, C., Wijesekera, R., & Bakker, M. G. (2021). Synthesis of mesoporous silica nanoparticles derived from rice husk and surface-controlled amine functionalization for efficient adsorption of methylene blue from aqueous solution. *Current Research in Green and Sustainable Chemistry*, 4, 100116. <https://doi.org/10.1016/j.crgsc.2021.100116>
- Waliullah, R. M., Rehan, A. I., Awual, M. E., Rasee, A. I., Sheikh, M. C., Salman, M. S., Hossain, M. S., Hasan, M. M., Kubra, K. T., Hasan, M. N., Marwani, H. M., Islam, A., Rahman, M. M., Khaleque, M. A., & Awual, M. R. (2023). Optimization of toxic dye removal from contaminated water using chitosan-grafted novel nanocomposite adsorbent. *Journal of Molecular Liquids*, 388, Article 122763. <https://doi.org/10.1016/j.molliq.2023.122763>
- Wang, X., Zhu, X., & Meng, X. (2017). Preparation of a Mg/Al/Fe layered supramolecular compound and application for removal of Cr(VI) from laboratory wastewater. *RSC Advances*, 7(56), 34984–34993. <https://doi.org/10.1039/c7ra04646d>
- Wijaya, A., Siregar, P. M. S. B. N., Badri, A. F., Palapa, N. R., Amri, A., Ahmad, N., & Lesbani, A. (2023). Modified layered double hydroxide Mg/M3+ (M3+ = Al and Cr) using metal oxide (Cu) as adsorbent for methyl orange and methyl red dyes. *Periodica Polytechnica Chemical Engineering*, 67(2), 173–184. <https://doi.org/10.3311/PPCh.21608>
- Yadav, B. S., & Dasgupta, S. (2022). Effect of time, pH, and temperature on kinetics for adsorption of methyl orange dye into the modified nitrate intercalated MgAl LDH adsorbent. *Inorganic Chemistry Communications*, 137, 109203. <https://doi.org/10.1016/j.inoche.2022.109203>
- Zhang, S., Lee, J., Li, B., Zhang, T., Baeck, S.-H., & Shim, S. E. (2024). Industrial solid waste-based layered double hydroxide composite with enhanced adsorption affinity to Congo Red, Acid Red, and Reactive Red. *Korean Journal of Chemical Engineering*, 41(3), 829–838. <https://doi.org/10.1007/s11814-024-00088-3>
- Zhou, L., Slaný, M., Bai, B., Du, W., Qu, C., Zhang, J., & Tang, Y. (2021). Enhanced removal of sulfonated lignite from oil wastewater with multidimensional MgAl-LDH nanoparticles. *Nanomaterials*, 11(4), Article art. no. 861. <https://doi.org/10.3390/nano11040861>

- Zubair, M., Daud, M., McKay, G., Shehzad, F., & Al-Harthi, M. A. (2017). Recent progress in layered double hydroxides (LDH)-containing hybrids as adsorbents for water remediation. *Applied Clay Science*, 143, 279–292. <https://doi.org/10.1016/j.clay.2017.04.002>
- Zubair, M., Yasir, M., Machovsky, M., Baig, N., Saood Manzar, M., Odeh, J. A. S., Hassan, K., & M., Hawari, A., Cevik, E., Al-Ejji, M. (2024). Photocatalytically active layered double hydroxide-PVDF composite membranes for effective

remediation of dyes contaminated water. *Chemosphere*, 364, 143169. <https://doi.org/10.1016/j.chemosphere.2024.143169>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.